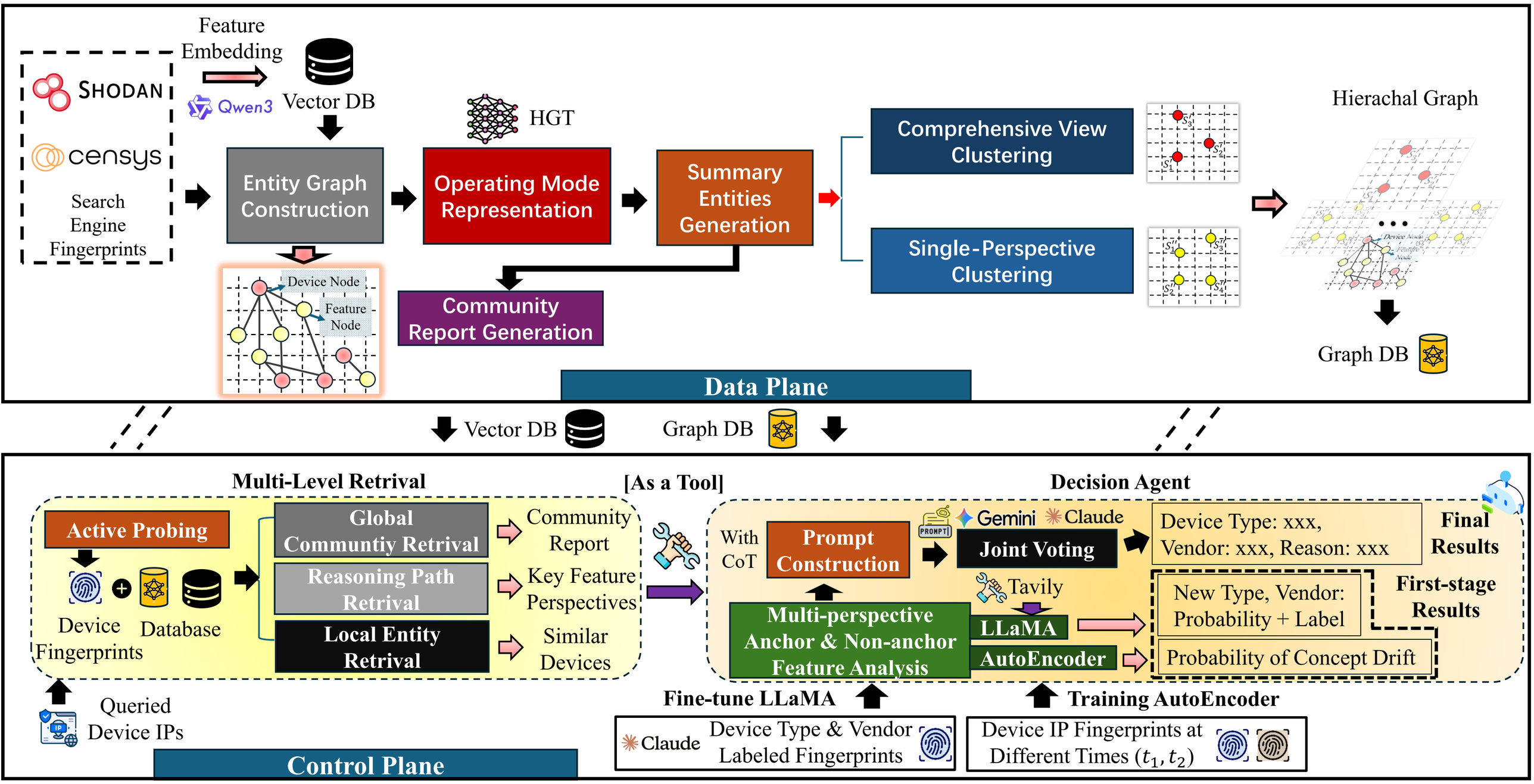
Feature Embedding
SHODAN
Qwen3
Vector DB
censys
Search Engine Fingerprints
Entity Graph Construction
Device Node
Feature Node
HGT
Operating Mode Representation
Summary Entities Generation
Community Report Generation
Comprehensive View Clustering
Single-Perspective Clustering
Hierachal Graph
Graph DB
Data Plane
Vector DB
Graph DB
Multi-Level Retrival
Active Probing
Device Fingerprints
Database
Global Communtiy Retrival
Reasoning Path Retrival
Local Entity Retrival
Community Report
Key Feature Perspectives
Similar Devices
[As a Tool]
Decision Agent
With CoT
Prompt Construction
Gemini
Claude
Joint Voting
Tavily
Multi-perspective Anchor & Non-anchor Feature Analysis
LLaMA
AutoEncoder
Device Type: xxx, Vendor: xxx, Reason: xxx
Final Results
New Type, Vendor: Probability + Label
First-stage Results
Probability of Concept Drift
Queried Device IPs
Control Plane
Fine-tune LLaMA
Claude
Device Type & Vendor Labeled Fingerprints
Training AutoEncoder
Device IP Fingerprints at Different Times (t_1, t_2)